

**Department of Electrical and Computer Engineering
National University of Singapore**

EE5508 Semiconductor Fundamentals (Semester I, AY2025/2026)

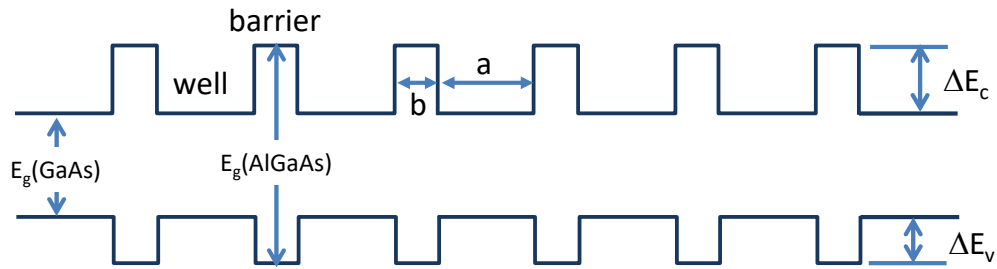
Assignment for Part I

Instructions:

- Choose **either one** of the following two assignments and submit a report by 19 October 2025 at 12:00 pm. Late submissions will not be accepted.
- The report should be uploaded to the 'Assignment Report for Part I' folder in Canvas. The file must be named in the following format: "Your name_ Matriculation Number.*".
- It is important to note that plagiarism is strictly prohibited and will result in a zero score for the assignment. All work must be your own, and proper citations must be provided for any references or sources used including generative AI.
- MATLAB installation (if needed):
 - a) Create your MATLAB account using your NUS email address:
<https://www.mathworks.com/mwaccount/>
 - b) Use the direct link to start the software installation:
<https://www.mathworks.com/academia/tah-portal/national-university-of-singapore-31340264.html>

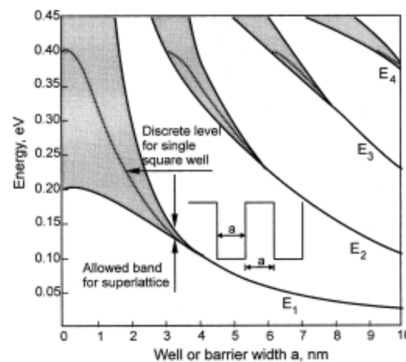
Assignment #1

Semiconductor superlattices (SLs) are artificial structures consisting of alternating layers of two semiconductors with similar lattice structure but different bandgaps. The most widely studied SL is GaAs/Al_xGa_{1-x}As in which both electrons and holes are confined in the GaAs layers (or potential wells) by the Al_xGa_{1-x}As barrier layers. When the barrier layer is sufficiently thick, overlapping of electron and hole wave functions between neighboring wells is negligible; therefore, the GaAs layers can be treated as isolated quantum wells. As a result, discrete energy levels are formed in the individual quantum wells. However, when the barrier thickness is reduced to allow significant mixing of wave functions from neighboring wells, instead of discrete energy levels, a continuous sub-band will be formed across the entire structure; hence it is called a superlattice. The depths of potential well for electrons and holes are determined by the conduction and valence band offsets, ΔE_c and ΔE_v , respectively. In the case of GaAs/Al_xGa_{1-x}As SL, it is found experimentally that 65% of the bandgap difference (ΔE_g) between GaAs and Al_xGa_{1-x}As goes to ΔE_c and the rest is taken up by ΔE_v . Bandgap of Al_xGa_{1-x}As is given by $1.424 + 1.247x$ eV, where x is the Al composition (you may ignore temperature dependence).



- Do a literature survey about the basic concept of superlattice and write a program to calculate the sub-band energy levels of GaAs/Al_{0.3}Ga_{0.7}As SL using the Kronig-Penney model. For the sake of simplicity, you may assume that the electron and hole effective masses of Al_{0.3}Ga_{0.7}As are the same as those of GaAs ($m_e^* = 0.067m_0$, $m_{hh}^* = 0.45m_0$, $m_{lh}^* = 0.082m_0$).
- Submit a report on your calculation results, which must include the sub-band energy level plots for electron, heavy hole and light hole as a function of well width (see example below). You may vary the well width continuously from 1 nm to 10 nm, and fix the barrier width at 5, 10, 20, 50, and 100 nm, respectively.
- Your report should be in the form of **MATLAB Live Scripts**.

For your reference, the calculated mini-bands for electrons may look like the following:



Example of calculated SL band structures for electrons

Assignment #2

The Fermi level in a non-degenerate Si can be found by solving the following equation:

$$N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) + \frac{N_a}{1 + 4 \exp\left(-\frac{E_F - E_a}{k_B T}\right)} = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right) + \frac{N_d}{1 + 2 \exp\left(-\frac{E_d - E_F}{k_B T}\right)}$$

Here, N_C (N_V) is the effective density of states in the conduction (valence) band, E_C (E_V) is the energy of conduction (valence) band edge, E_F is the Fermi energy, N_d (N_a) is the donor (acceptor) concentration, E_d (E_a) is the donor (acceptor) energy level, T is the temperature, and k_B is the Boltzmann constant. Once the Fermi level is known, the electron and hole density can be calculated as follows:

$$n = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \quad \text{and} \quad p = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

Develop an **Excel Spreadsheet** for calculating the Fermi level and carrier density (including both electrons and holes) of Si by treating N_d and N_a as input parameters. Plot carrier density (in log scale) and Fermi energy (in linear scale) as a function of reciprocal temperature ($1000/T$) over the range of $T = 10 - 800$ K. You are required to produce a total of three plots:

- i) $\log(n) - 1000/T$
- ii) $\log(p) - 1000/T$
- iii) $E_F - 1000/T$

The Excel Spreadsheet must be capable of calculating E_F , n and p , and updating the results automatically whenever N_d and N_a are changed. Submit the completed Excel Spreadsheet as the report for this assignment.

You may assume that the donor and acceptor impurities are phosphorus (P) and aluminum (Al), respectively. The following equations and parameters may be used for the calculations:

- a) $E_g = 1.17 - 4.73 \times 10^{-4} T^2 / (T + 636)$ (T : temperature in unit of Kelvin)
- b) $N_C = 2 \left(\frac{m_{de} k_B T}{2\pi \hbar^2} \right)^{3/2} M_C$, $N_V = 2 \left(\frac{m_{dh} k_B T}{2\pi \hbar^2} \right)^{3/2}$
- c) $m_{de} = (m_1^* m_2^* m_3^*)^{1/3}$, $m_{dh} = (m_{lh}^{*3/2} + m_{hh}^{*3/2})^{2/3}$
- d) $m_1^* = m_2^* = 0.19 m_e$, $m_3^* = 0.98 m_e$, $m_{lh}^* = 0.16 m_e$, $m_{hh}^* = 0.49 m_e$, $M_C = 6$
- e) $m_e = 9.109 \times 10^{-31} \text{ kg}$
- f) Ionization energy of phosphorous in Si: 0.045 eV
- g) Ionization energy of aluminum in Si: 0.067 eV

For your reference, for P-doped Si with $N_d = 10^{15} \text{ cm}^{-3}$, $N_a = 0$, the $\log(n) - 1000/T$ plot may look like the following:

